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CogECI: Context Grounded Document-level Event Causality Identification via Large Language Models

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ABSTRACT

Document-level Event Causality Identification (ECI) focuses on recognizing causal relationships between pairs of events in a document. Previous approaches mainly capture implicit causal links by directly utilizing pre-trained language models to analyze the text. However, they may rely on explicit language shortcuts, leading to predictions with weak correlations between event causality and their grounded context. To tackle this challenge, in this paper, we propose a novel **Co**ntext **G**rounded Document-level **E**vent **C**ausality **I**dentification (CogECI) Network. Specifically, we first utilize Large Language Models (LLMs) to extract the grounded context pseudo-labels from the document that are semantically contributive to the causality reasoning of the event pairs. Subsequently, the grounded context pseudo-labels are utilized for supervised Context Grounding Learning (CGL) with the Sentence Graph Module (SGM) and Dual-Stream Causality Reasoning Module (DSCRM) to eliminate the bad influence of numerous redundant and noisy long contexts. Extensive experiments on two benchmark datasets show that CogECI not only greatly outperforms previous state-of-the-art (SOTA) methods but also leads to precise context grounding ability. Our code will be available soon.

1. Introduction

Document-level Event Causality Identification (ECI) [1–3] aims to capture causal relationships among events in a document. It is a crucial task in Natural Language Processing (NLP) that can enable more advanced applications, such as information extraction [4] and question answering [5,6]. Although methods [7–10] based on Transformers or Graph Neural Networks (GNNs) could establish global dependencies between events and achieve significant progress, challenges remain in achieving precise and reliable Event Causality Identification due to limitations from both data and model aspects:

1. In terms of data. Document-level event causality identification lacks explicit causal indicators such as "because" or "caused by", necessitating models to extend beyond event-centric dependencies and establish comprehensive context dependencies across multiple sentences for multi-hop reasoning and coreference resolution. Specifically, as shown in Fig. 1, when identifying the causal relation between "afraid" and "a magnitude 9.1 quake", the model should first pay attention to the people's region "Bener Meriah". Then, ground the critical context that "Bener Meriah was one of the worst hit areas" and the reason is "quake". Additionally, resolve the coreference between "quake" and "a magnitude 9.1 quake". Finally, the model could accurately interpret their causal relation. According to this example, we could find that addressing multi-hop reasoning and coreference resolution must extract and align implicit contextual clues related to the target events. These contextual clues are not limited to the intermediate event-centric reasoning [11–13]. More importantly, even non-event contexts, which lack explicit connections to the target events, should be considered, as they play a crucial role in building a coherent and meaningful understanding of event causality.

2. In terms of model. Strong ECI methods primarily rely on powerful language models, such as BERT [14] and RoBERTa [15], for text encoding. While these pre-trained models exhibit robust semantic understanding capabilities, their dependence on this generalized understanding ability can lead to overconfidence in the knowledge acquired during the pre-training stage. Some studies propose to use the diffusion method [16] to remove redundant context, or adopt graph reasoning structures [12,13] to infer the associations between context and labels, as well as using event cloze tasks [17] to enhance the understanding of relations among events. Although these methods achieve excellent progress, they are still limited by the knowledge reserves and event comprehension capabilities. Moreover, they are typically trained with

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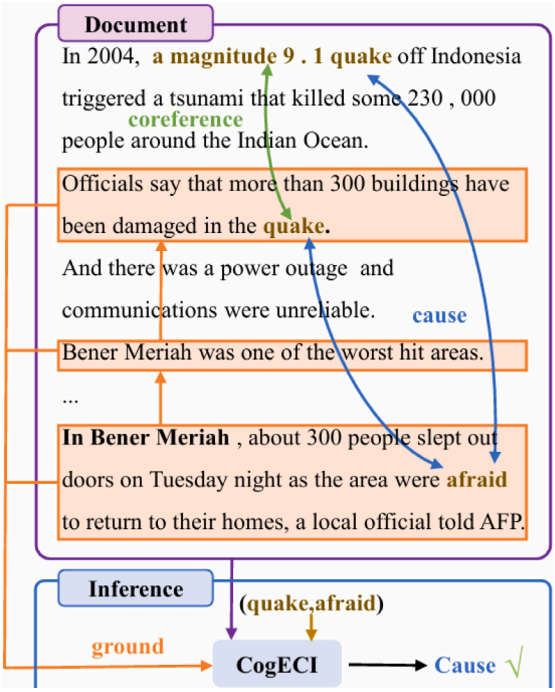


Fig. 1. Example of the dependencies among sentences in reliable event causality identification.

purely supervised learning approaches designed specifically for causality identification. As a result, these models often tend to develop a reliance on spurious correlations between inputs and target labels. This hinders their ability to accurately capture the precise causal relations between event pairs and their grounded contextual clues, which are essential factors for effective generalization. Consequently, such models are highly susceptible to overfitting.

To achieve context-grounded and robust event causality identification, we focus on improving the model's ability to perform relevant contextual clues grounding during the event reasoning process. Regarding the limitations mentioned above, we leverage Large Language Models (LLMs) for their extensive commonsense knowledge and advanced semantic understanding capabilities to analyze the documents and extract context that is relevant for reasoning the event relationships. This enables the model to obtain a more comprehensive and accurate event relation understanding. Based on these data, we further introduce a novel **Co**ntext **G**rounded Document-level **E**vent **C**ausality **I**dentification (CogECI) model, which emphasizes Event Causality Learning (ECL) and Context Grounding Learning (CGL). CogECI starts with a Sentence Graph Module (SGM) to establish global sentence-level dependencies, addressing the limitations of traditional sliding window approaches that fail to capture global semantic relations across the document. This also facilitates the grounding of relevant contextual clues. Next, events enriched with global dependencies are extracted and fed into the Event Causality Reasoning Module. Our research reveals a significant imbalance in the number of grounded sentences between inter-sentence and intra-sentence causality identification. Using a shared module for both leads to a model bias toward inter-sentence causality. To mitigate this, we decouple the two levels of Event Causality Identification and introduce a Dual-Stream Causality Reasoning Module (DSCRM) to handle them separately. Finally, the event reasoning result is extracted and used for Event Causality Learning and Context Grounding Learning. For CGL specifically, we compute the context grounding loss based on the reasoning results from DSCRM and the generated grounded context. This label is used only during training, serving as a form of supervision to equip the model with the ability to distinguish and align relevant context. The experimental results demonstrate that our method outperforms previous approaches in EventStoryLine and MAVEN-ERE.

Our contributions can be summarized as follows:

- We highlight the importance of implicit contextual clues in Event Causality Identification, and propose a novel method for grounded context extraction and context grounding learning using Large Language Models.
- We propose a Context Grounded Document-level Event Causality Identification Network with a Sentence Graph Module, and a Dual-Stream Causality Reasoning Module for context grounding learning and tackling the imbalance issue between inter-sentence and intra-sentence reasoning.
- Extensive experiments on two benchmark datasets (EventStoryLine and MAVEN-ERE) validate the effectiveness of CogECI.

2. Related work

Document-level Event Causality Identification has gained increasing attention, though it remains challenged by noisy contexts, long-range discourse, and implicit reasoning [18].

2.1. Deep semantic encoding based document-level ECI

These methods enhance contextual representation through advanced neural architectures or noise-robust learning strategies. SemDI [17] employs a Cloze Analyzer to query causal signals from a comprehensive context. More recently, HOTECI [19] introduces a hierarchical learning paradigm that dynamically identifies and amplifies causally salient spans across multiple granularities. Meanwhile, DiffusECI [16] reframes causal representation learning as a diffusion-like process, modeling gradual transitions from noisy context to clean causal labels to explicitly suppress irrelevant information. ICCL [20] adopts contrastive learning to improve the representation quality of both positive and negative examples. DPJL [21] exploits the implicit and explicit causal information contained in pre-trained language models to mitigate data scarcity. GenECI [22] captures the event semantic dependencies within the context in the pre-trained language models with multi-task learning.

2.2. Event graph reasoning based document-level ECI

LSIN [23] incorporates external knowledge (e.g., ConceptNet, COMET) to build descriptive and relational graphs for enhanced reasoning. ERGO [24] designs a relational graph framework to alleviate false positives/negatives by explicitly modeling event interactions. Furthermore, CHEER [13] introduces centrality-aware graph propagation to preserve core events that span the entire document, treating peripheral events as their causes or effects. SemSIn [11] integrates event-level structural cues through a GNN-driven event aggregator and models inter-event structural dependencies with an LSTM-based path aggregator. iLIF [12] leverages structural information and causal directionality within graph representations to refine event embeddings for downstream identification (see Fig. 2).

2.3. Large language models based document-level ECI

With the rapid advancement of large language models, their powerful language understanding and vast knowledge reserves have brought significant progress to traditional methods, offering new perspectives and capabilities in Event Causality Identification. LKCER [25] utilizes COMET-generated knowledge, enriched through LLMs, to strengthen the relationships among event pairs and the relevant knowledge. To enhance the causal reasoning capabilities of LLMs, Dr.ECI [26] effectively decomposes the intricate ECI task and utilizes the causal structure

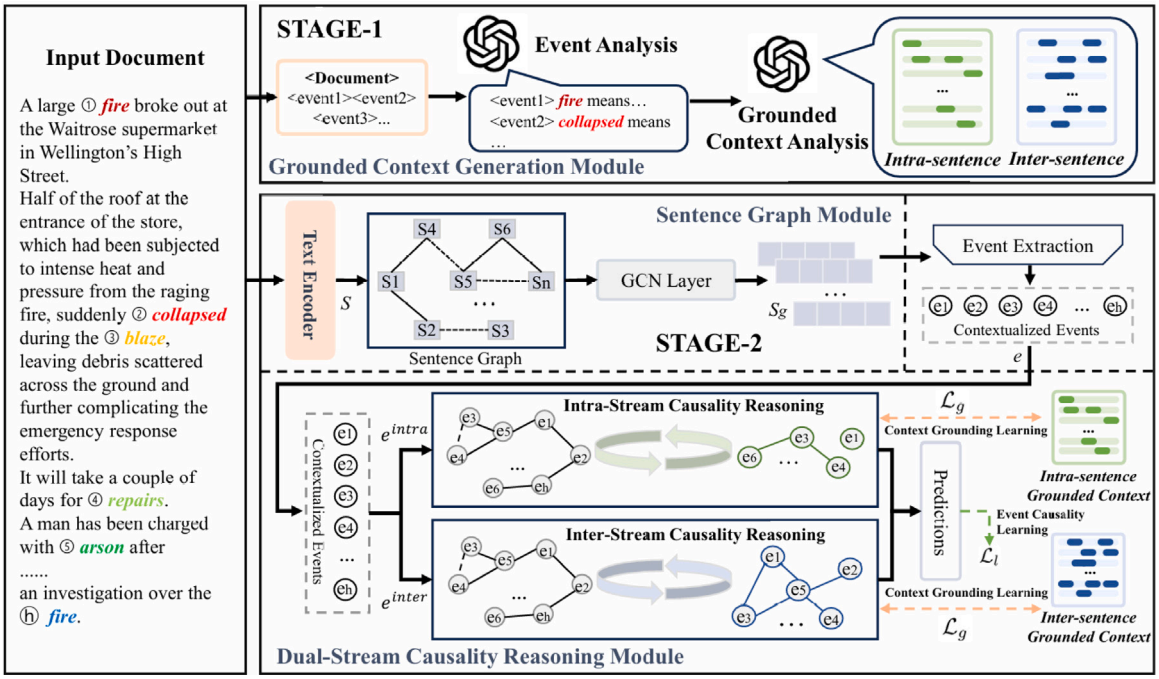


Fig. 2. The framework of Context Grounded Document-level Event Causality Identification model. Stage 1: Labeling the grounded context based on prompt learning with Large Language Models. Stage 2: Event Causality Learning and Context Grounding Learning based on our proposed Context Grounded Document-level Event Causality Identification Network, which contains a Sentence Graph Module and a Dual-Stream Causality Reasoning Module.

knowledge for addressing implicit causality and indirect causality challenges. Chen et al. [27] investigate three distinct strategies to equip LLMs with event relational reasoning, thereby improving their ability to produce coherent responses in diverse contexts. MRBalance [28] leverages role-based multi-agent debates and optimizes causal reasoning across diverse scenarios. Xiang et al. [29] propose an Instructively Generation and Statement Evaluation method to identify both the existence and direction of causality. Additionally, for solving the causal hallucinations, KnowQA [28] performs fine-tuning of LLMs in two stages: Event Structure Construction and Binary Question Answering. MEFA [30] uses Multi-source Evidence Fuzzy Aggregation to decompose causality reasoning tasks and effectively reduce causal hallucinations.

Although the aforementioned approaches have achieved promising results, the large language models based methods may lead to substantial computational costs. Deep Semantic Encoding and Event Graph Reasoning based ECI models still struggle to filter irrelevant contextual noise and establish robust grounding for global causal chains. To address this, we propose leveraging Large Language Models (LLMs) to guide the model in learning critical context, reducing noisy interference, and promoting the correct context understanding of global causal relations.

3. Methodology

As show in Fig. 2, our method contains two stages: (1) labeling the grounded context based on prompt learning with Large Language Models; (2) Event Causality Learning and Context Grounding Learning based on our proposed Context Grounded Document-level Event Causality Identification Network, which contains a Sentence Graph Module and a Dual-Stream Causality Reasoning Module. The Sentence Graph Module aims to establish global sentence-level dependencies, addressing the limitations of traditional sliding window approaches that fail to capture global semantic relations across the document. The Dual-Stream Causality Reasoning Module performs intra-sentence and inter-sentence causality identification, respectively.

3.1. Grounded context generation

Grounded context requires strong semantic understanding ability and a commonsense knowledge reserve. To extract this context, we apply powerful LLMs (e.g., Qwen2.5 [31], Llama-3-8b [32], ChatGPT-4o-mini [33]) for the document analysis. Specifically, we first prompt the LLMs with prompt P_m to analyze the event meaning according to the document D. Next, use prompt P_g to guide the LLM in performing grounded context analysis, identifying which sentences contain critical connecting information GC necessary for accurately assessing the relation between event E_i and event E_j as follows:

$$Event_{detail} = LLM(P_m, E_i, E_j, D), \tag{1}$$

$$GC = LLM(P_g, E_i, E_j, Event_{detail}, D). \tag{2}$$

Specifically, the prompts P_m and P_g are defined as follows:

P_m: Please fully understand the events E_i and E_j in the document D and generate a subject–verb–object short sentence to describe these events.

P_g: The input document D is labeled with an ordinal number before each sentence, e.g., (0,1...). Knowing that events: $\{E_i \& Event_i^{detail}\}$, $\{E_j \& Event_j^{detail}\}$, in the document are causally related, please identify which sentences from the document articulate the causal relationship between events E_i and E_j. Answer the related sentence numbers as a list, e.g., "[1,2, …]".

Notably, when generating grounded context, we address both inter-sentence and intra-sentence issues simultaneously. This is because, in intra-sentence scenarios, a single sentence often lacks sufficient information for robust reasoning, making it essential to incorporate additional context to support accurate inference. Further analysis of the grounded context is provided in Section 4.5.

3.2. Text encoder

Given a document D with n sentences and h events. The wth sentence contains m words. We utilize the BERT-base model to encode each sentence and get the encoding outputs $S = \{s_1, s_2, \ldots, s_n\}$.

3.3. Sentence graph module

Due to the limited processing length of BERT, encoding documents can only be performed on individual sentences separately, resulting in the loss of document-level semantic understanding. Although there is a dynamic window method for multi-sentence dependency modeling, it cannot catch the global long-distance dependency. Moreover, the context grounding learning requires inter-sentence relational priors for subsequent reasoning. To address this, we introduce the sentence graph module to model the overall semantic relation and capture the contextual clues within the document. Considering the number of sentences, we employ a Graph Convolution Network to model the global relations among the sentences.

Specifically, we first construct the sentence graph to capture the semantic relations between multiple sentences. Its adjacency matrix R_s is computed through pairwise similarities as follows:

$$R_s = softmax(W_k(S)W_v(S)^T). \tag{3}$$

Here, T means the matrix transpose. $W_k()$, $W_v()$ are linear operations. Then, we apply the U-layer graph attention convolution layers to capture the global relations,

$$S^{u+1} = ReLU((R_s + I)S^u P^u), \tag{4}$$

where P^u is the graph parameters of the uth layer. I is the identity matrix for skip connections. S^u is initialized by the output node representations S. S_g is the node reasoning result. Subsequently, we represent the ith event as e_i with the average of individual token representations of the event trigger words from the sentence feature S_g.

3.4. Dual-stream causality reasoning module

The grounded sentences in the intra-sentence scenario are located nearby and involve relatively fewer sentences, whereas in the inter-sentence scenario, they often span larger distances between sentences, resulting in a significantly higher number of grounded sentences being involved. Therefore, there remains a significant difference in the grounded context required for these two types of samples, and we place these two types of issues into separate branches for graph reasoning.

In this module, we adopt the Graph Attention Networks (GAT) instead of the GCN due to its ability to assign adaptive attention weights, which is more suitable for capturing the varying importance of event interactions. More details are discussed in Section 4.5.

We begin by initializing a two-level graph representation $\mathcal{N}^{inter}$ and $\mathcal{N}^{intra}$, which contain the event nodes e and their relations $\mathcal{E}$. Next, we compute the importance coefficients c_{ij} for different graphs using a shared attention mechanism.

$$c_{ij} = softmax\left(\frac{e_i e_j}{\sqrt{d}}\right)e_j, \tag{5}$$

where d means the dimension of e_j. We next compute the weight for the edge $\mathcal{E}_{ij}$ which means the edge from e_i to e_j as follows:

$$\mathcal{E}_{ij} = softmax(c_{ij}). \tag{6}$$

Then, we update the causal graph representation for the event node by the following multi-head attention mechanism:

$$\mathbf{e}_i^{inter} = \Big\|_{k=1}^{K} \sigma\left(\sum_{j\in\mathcal{N}_i^{inter}} \mathcal{E}_{ij}^{inter}\mathbf{W}^k e_i\right), \tag{7}$$

$$\mathbf{e}_i^{intra} = \Big\|_{k=1}^{K} \sigma\left(\sum_{j\in\mathcal{N}_i^{intra}} \mathcal{E}_{ij}^{intra}\mathbf{W}^k e_i\right), \tag{8}$$

where $\sigma()$ denotes an element-wise activation function and $\|$ means the concatenation operation. K means the number of the graph layers.

Finally, we concatenate the contextual text representation e with the graph representation $\mathbf{e}$ and apply a multi-layer perception to get the fusion result p_{ij} for event causality learning:

$$p_{ij} = Softmax(MLP(e_i \| e_j \| (\mathbf{e}_i - \mathbf{e}_j))). \tag{9}$$

This fusion process of events is followed by iLIF [12].

3.5. Training strategy

We adopt the α-balanced variant of focal loss to address the issue of class imbalance and compute the event causality classification loss $\mathcal{L}_l$ as follows:

$$\mathcal{L}_l = -\sum_{e_i,e_j\in D} \alpha^c (1 - p_{ij}^c)^\gamma \log(p_{ij}^c), \tag{10}$$

where p_{ij}^c is the predicted probability of the true class c, α^c is a weighting factor for the true class c. Additionally, γ represents a predefined focusing hyper-parameter.

For the Context Grounding Learning, based on the generated grounded context GC, we assign a binary label to each sentence in the document: if the current sentence appears in GC, it is assigned a value of 1; otherwise, it is assigned 0. This forms a binary label vector g_{label} which also refers to the "grounded context pseudo label". Unlike previous methods that focus solely on predicting the event relation label, our proposed CogECI leverages grounding pseudo labels based on the grounded context. This design guides the model not only to predict the correct relation but also to accurately identify the key supporting sentences. As a result, the model achieves a deeper understanding of the relation while simultaneously improving interpretability through explicit evidence grounding. Specifically, for each sentence, we compute the grounding score g_{score} to the current event pair with the cosine similarity equation and apply a softmax function over all scores, as follows:

$$\mathbf{e}_{pair} = Linear(\mathbf{e}_i \| \mathbf{e}_j), \tag{11}$$

$$g_{score} = Softmax(cosine_similarity(\mathbf{e}_{pair}, S)). \tag{12}$$

A binary prediction matrix is then obtained by thresholding these scores: if a sentence's score is above the mean, it is set to 1; otherwise, it is set to 0.

$$g_s = \begin{cases} 1, & \text{if } g_{score} > 1/n \\ 0, & \text{otherwise,} \end{cases} \tag{13}$$

where n means the sentence number in the document. Then, the predicted binary matrix g_s is compared with the label vector g_{label} using binary cross-entropy (BCE) loss to compute the multi-label classification loss, as follows:

$$\mathcal{L}_g = -\frac{1}{N}\sum_{e_i,e_j\in D} \left[g_{\text{label}} \cdot \log(g_s) + (1 - g_{\text{label}}) \cdot \log(1 - g_s)\right], \tag{14}$$

where N indicates how many event pairs are present in the document.

Finally, the two losses are summed to form the final loss $\mathcal{L}$ as follows:

$$\mathcal{L} = \mathcal{L}_l + \beta * \mathcal{L}_g, \tag{15}$$

where β is a hyper-parameter which is analyzed in Section 4.5.

4. Experiments

This section reports comprehensive experimental results on two datasets. Our overarching goal is to answer the following research questions:

- RQ1: How does our model compare with the state-of-the-art Document-level ECI approaches on quantitative results?

Table 1
Comparison of results on the EventStoryLine dataset, with the best-performing metrics shown in bold and the second-best underlined. Event pairs labeled as "Intra-sentence" appear in the same sentence, whereas "Inter-sentence" pairs are distributed across different sentences.

Model	Intra-sentence			Inter-sentence			Intra+Inter		
	Precision (%)	Recall (%)	F1 (%)	Precision (%)	Recall (%)	F1 (%)	Precision (%)	Recall (%)	F1 (%)
Large Language Models									
text-davinci-003 [34]	33.2	74.4	45.9	–	–	–	–	–	–
GPT-3.5-turbo [17]	39.5	40.3	39.7	–	–	–	–	–	–
GPT-4 [17]	30.7	**85.7**	45.2	–	–	–	–	–	–
LLaMA2-7B [17]	11.4	50.0	18.6	–	–	–	–	–	–
LLaMA2-7B-FT [17]	20.5	57.1	29.8	–	–	–	–	–	–
Normal models									
BERT [14]	60.4	45.7	52.0	30.6	39.1	34.3	37.2	41.2	39.1
RoBERTa [15]	62.7	45.4	52.7	32.7	38.3	35.3	39.7	40.6	40.1
LONG [35]	47.7	69.3	56.5	26.1	55.6	35.5	31.4	60.0	41.2
KnowBERT [36]	41.9	32.5	50.1	–	–	–	–	–	–
KnowDis [37]	39.7	66.5	49.7	–	–	–	–	–	–
CauSeRL [38]	41.9	69.0	52.1	–	–	–	–	–	–
GenECI [22]	59.5	57.1	58.8	–	–	–	–	–	–
ERGO [24]	49.7	72.6	59.0	43.2	48.8	45.8	46.3	50.1	48.1
DPJL [21]	65.3	70.8	67.9	–	–	–	–	–	–
PPAT [39]	62.1	68.8	65.3	54.0	50.2	52.0	56.8	56.0	56.4
CHEER [13]	56.9	69.6	62.6	45.2	52.1	48.4	49.7	53.3	51.4
SENDIR [40]	65.8	66.7	66.2	33.0	**90.0**	48.3	37.8	**82.8**	51.9
SemSIn [11]	50.5	63.0	56.1	–	–	–	–	–	–
ICCL [20]	64.9	69.6	67.1	56.3	58.4	57.2	59.0	61.9	60.4
SemDI [17]	56.7	68.6	62.0	–	–	–	–	–	–
iLIF [12]	76.8	66.3	71.2	53.5	65.9	59.1	59.2	66.1	62.5
HOTECI [19]	66.1	72.3	69.1	**81.4**	40.6	55.1	63.1	51.2	56.5
LKCER [25]	67.3	72.7	69.8	–	–	–	–	–	–
DiffusECI [16]	65.3	78.3	71.4	61.9	59.9	**60.9**	63.0	64.1	63.5
CogECI (Ours)	**80.6**	76.1	**78.3**	66.5	55.6	60.6	**71.4**	62.2	**66.5**

- RQ2: How much do various components of our model contribute to its overall performance?
- RQ3: How much do various LLMs and different hyper-parameters contribute to the overall performance of CogECI?
- RQ4: How does the proposed model compare to other methods in the case study?

4.1. Experimental setup

Datasets Details: We evaluate our CogECI model using two widely adopted datasets. The EventStoryLine dataset (v0.9) [41] consists of 22 topics, 258 documents, and 5334 events, of which 1770 intra-sentence and 3885 inter-sentence event pairs are annotated for causal relations. In line with Gao et al. [42], we organize the documents by topic, using the last two topics as development data and the other 20 for 5-fold cross-validation.

Another dataset is MAVEN-ERE [43], which comprises 4480 documents, 103,193 events, and 57,992 causal event pairs. We select the 'CAUSE' and 'PRECONDITION' types to represent CAUSE relations. As the setting of iLIF [12], we use the original development set as the test set, then sample 10% of the data from the original training set to form the development set. However, the current best model iLIF [12] does not release the detail split val set, and different val sets will cause different performances. Therefore, we tried their code and chose the best performance for our experiment setting for fair comparison.

Additionally, in these two datasets, Intra-sentence refers to event pairs occurring within the same sentence, while Inter-sentence refers to event pairs occurring across different sentences. The Intra+Inter setting reports the overall performance containing both types of event pairs.

Implementation Details: Our method is implemented using the PyTorch version of the Hugging Face Transformers library. We adopt the uncased BERT-base model [14] as the underlying pre-trained language model and fine-tune it during training. The layer number U of the graph attention convolution is set to 2 in the sentence graph module. The graph layer number K in the Dual-Stream Causality Reasoning Module is set with the iteration method in iLIF [12], which is terminated when the structural difference between the last and current graph is less than 2. The weighting factor of CAUSE class α is set to 0.75, and the focusing parameter γ to 2. The model uses AdamW for optimization, including a linear warm-up over the first 10% of steps. Learning rates are 2e−5 for the PLM and 1e−4 for other modules, with batch size 1. Training is carried out on a 48 GB NVIDIA A40 GPU.

Evaluation Metrics: Following others [13], the model is assessed using Precision, Recall, and F1-score as evaluation metrics.

4.2. Baselines

Large Language Model based methods include works [17,34] like Text-davinci-003, GPT-3.5-turbo, GPT-4 [33], and LLaMA-2-7B [32], which use prompt-based techniques. LLaMA-2-7B-FT is fine-tuned with ECI datasets. GenECI [22] focuses on multi-task learning to generate dependency paths between event mentions for ECI.

KnowBERT [36] integrates external commonsense knowledge and a mention masking technique. KnowDis [37] designs a knowledge-enhanced distant data augmentation approach. CauSeRL [38] designs a self-supervised method with external causal statements. GenECI [22] employs a generative approach to jointly generate causal links and dependency paths between input event mentions, optimized via reinforcement learning. ERGO [24] builds a complete event relational graph without external tools or prior knowledge. DPJL [21] leverages pre-trained causal knowledge to enhance the identification of explicit and implicit causality. PPAT [39] applies a progressive reasoning approach, first predicting intra-sentence causality, then inferring inter-sentence causality. CHEER [13] strengthens important causal structures and central events. SENDIR [40] classifies event pairs as either within the same sentence or across sentences by measuring their information density. SemSIn [11] employs a GNN-based aggregator to incorporate event-centric structural information. ICCL [20] applies contrastive learning to improve the effectiveness of causal demonstrations. SemDI [17] captures semantic dependencies with a unified encoder and a Cloze Analyzer for causal relation understanding. iLIF [12] emphasizes using causality directionality and structure to improve event representations

Table 2
Comparison of results on the MAVEN-ERE dataset. The iLIF* means the results reported by us, and the experiment setting of our results is the same as iLIF*.

Model	Intra-sentence			Inter-sentence			Intra+Inter		
	Precision (%)	Recall (%)	F1 (%)	Precision (%)	Recall (%)	F1 (%)	Precision (%)	Recall (%)	F1 (%)
Large Language Models									
text-davinci-003 [34]	25.0	75.1	37.5	–	–	–	–	–	–
GPT-3.5-turbo [17]	19.9	85.8	32.3	–	–	–	–	–	–
GPT-4 [17]	22.5	**92.4**	36.2	–	–	–	–	–	–
Normal models									
BERT [14]	46.8	50.3	48.5	43.0	46.8	44.8	43.3	47.1	45.1
ERGO [24]	63.1	65.3	64.2	48.7	**62.0**	54.6	49.6	**62.3**	55.2
SENDIR [40]	51.4	53.6	52.5	51.9	52.8	52.4	51.9	52.9	52.4
PPAT [39]	37.9	66.7	47.7	32.6	39.3	35.6	34.2	46.2	39.2
iLIF* [12]	**74.0**	51.7	60.8	**64.7**	47.8	54.9	**65.4**	48.1	55.4
iLIF [12]	74.4	51.5	60.9	67.1	49.2	56.8	67.7	49.4	57.1
CogECI (Ours)	69.0	58.6	**63.4**	61.8	55.5	**58.5**	62.4	55.8	**58.9**

for future identification. HOTECI [19] incorporates a new learning paradigm to enhance important context extraction. LKCER [25] introduces the event conceptual information and rich event knowledge to enhance the representations of event pairs. DiffusECI [16] refines causal representations by modeling gradual transitions between context and causal labels to reduce noise.

4.3. Quantitative results (RQ1)

EventStoryLine dataset: Table 1 presents the overall performance comparison between our CogECI model and the current baseline models on the EventStoryLine dataset. The F1 score provides a more comprehensive measure of the models' overall effectiveness.

From the results, we observe that intra-sentence reasoning also relies heavily on accurate contextual dependencies. CogECI yields the most noticeable improvement (intra F1 +6.9%) compared with DiffusECI. This is mainly because intra-sentence reasoning requires relatively fewer grounded contexts. However, class imbalance often hinders the model's ability to effectively learn precise contextual dependency information. By introducing grounded context learning, our model achieves significant improvements.

For inter-sentence reasoning, despite the availability of abundant contextual relations, our model achieves comparative performance in terms of precision. However, recall remains relatively low. This is primarily due to the complexity of inter-sentence causal reasoning: the contexts generated by LLMs often involve intricate, overlapping contexts, making it challenging for the model to infer complex relations accurately. Thus, the improvement in this aspect is relatively limited compared with the DiffusECI.

When analyzed with intra+inter, our method demonstrates a significant advantage in causal relation identification, outperforming the second-best model by 3.0% in intra+inter F1 score. Moreover, our approach still significantly enhances the model's precision (P +8.4%). This improvement primarily stems from transferring the long context understanding ability and event-related grounding knowledge from LLMs to CogECI for reasoning about causal relations between events. Nevertheless, the Recall is relatively poor (−3.9%) than iLIF. It is primarily due to the complexity of inter-sentence causal reasoning as analyzed above.

Additionally, compared to LLM-based methods, whether in zero-shot or fine-tuning settings, our approach consistently achieves superior results. This demonstrates that our method successfully complements small models by leveraging the strengths of Large Language Models.

MAVEN-ERE dataset: Similarly, Table 2 shows the overall performance comparison between our CogECI model and the current baseline models on the MAVEN-ERE dataset. Since this dataset is relatively new, fewer experiments have been conducted on it, and the validation set splits vary, leading to significant performance fluctuations and potentially unfair comparisons. To ensure fairness, we first reproduced the iLIF [12] model and optimized its performance as much as possible. Based on this setup, we conducted experiments using the same configuration. Experimental findings demonstrate that our method reaches leading performance (intra+inter F1 +3.5%), confirming the generalizability of our approach.

Table 3
Ablation study on EventStoryLine. CGLoss means the Context Grounding loss, SGM means Sentence Graph Module, and DSCRM means Dual-Stream Causality Reasoning Module.

Model	Intra+Inter		
	Precision (%)	Recall (%)	F1 (%)
CogECI	**71.4**	62.2	**66.5**
CogECI w/o CGLoss	70.1	58.2	63.4
CogECI w/o SGM	67.3	60.6	63.3
CogECI w/o DSCRM	63.9	**63.4**	63.5

4.4. Ablation study (RQ2)

The ablation experimental results for CogECI are presented in Table 3. We conduct ablation studies to examine the impact of each component in our proposed framework. The full model CogECI incorporates the Dual-Stream Causality Reasoning Module, the Sentence Graph Module (SGM), and the Context Grounding loss (CGloss). When removing the context grounding loss, we observe a consistent performance drop, indicating that grounding contextual clues plays a crucial role in improving causal reasoning. Removing the Dual-Stream Causality Reasoning Module (DSCRM) and instead using a shared graph module for both intra- and inter-sentence reasoning also leads to a noticeable decline in performance. This supports our finding that the distinct context needs of intra- and inter-sentence causality require separate modeling. Finally, without the Sentence Graph Module, the model struggles to capture global inter-sentence dependencies effectively, resulting in the most significant drop in performance. These results highlight the importance of each component, especially in enabling the model to represent and reason over contextual dependencies more accurately.

4.5. Parameter study (RQ3)

In Fig. 3, we compare the grounded context generated by different LLMs, including GPT-4o-mini, Llama-3-8b, Qwen-2.5-7b, and Llama3-70b. It can be observed from the figure that the context generated by Llama3-8b is more accurate, leading to the best performance of the

Table 4
Different types of threshold design in Eq. (13), where n means the sentence number in the document.

Threshold	Intra+Inter		
	Precision (%)	Recall (%)	F1 (%)
$1/n$	**71.4**	**62.2**	**66.5**
0.1	70.4	61.9	65.8
0.5	70.8	62.0	66.1

Table 5
Different graph design in the Dual-Stream Causality Reasoning Module.

Model	Intra+Inter		
	Precision (%)	Recall (%)	F1 (%)
Graph Attention Network	71.4	**62.2**	**66.5**
Graph Convolutional Network	**72.2**	60.9	66.0
HypergraphConv	72.2	60.2	65.7
TransformerConv	70.75	61.7	65.9

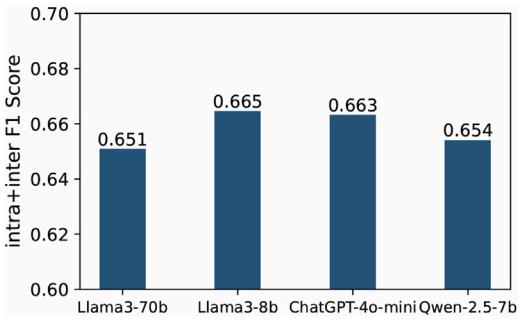


Fig. 3. The parameter study of different LLMs.

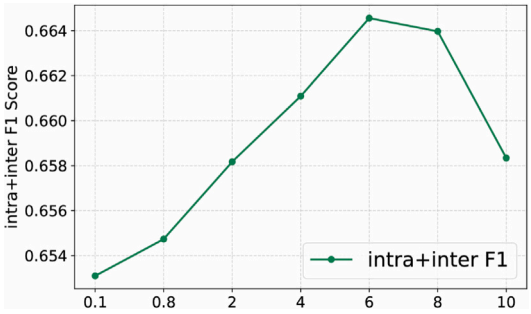


Fig. 4. The parameter study of CogECI with β.

model. In addition, we annotated 200 grounded contexts generated by Llama3-8b and GPT-4o-mini to evaluate their accuracy. The results (Completely Correct: Llama3-8b 57% GPT-4o-mini 34%, Incomplete but Correct: Llama3-8b 21% GPT-4o-mini 32%, Incorrect: Llama3-8b 22% GPT-4o-mini 34%) indicate that Llama3-8b demonstrates higher accuracy in the generated grounded context, which also highlights the significance of the context grounding learning. Stronger language models may over-analyze and incorporate irrelevant information, ultimately compromising performance. This phenomenon partly explains why the LLaMA3-70B model can underperform compared to its 7B counterpart in ECI.

In Fig. 4, an analysis is conducted on the hyper-parameter β for adding context grounding loss and event causality classification loss. It can be seen that the model achieves optimal results when β is set to 6.

In Fig. 5, we analyze the number U of graph attention convolutions in the Sentence Graph Module. We could see that the model achieves optimal results when U is set to 2.

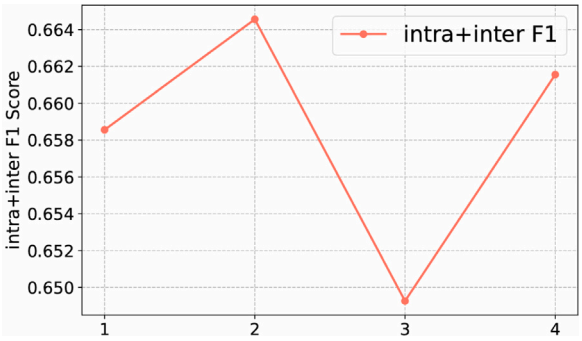


Fig. 5. The parameter study of the number of GCN U.

In Table 4, we analyze the different types of the threshold in Eq. (13). From the results, it can be seen that the dynamic threshold set based on sentence length n is better than the fixed static threshold method.

Additionally, in Table 5, we analyze several graph-based methods used in the Dual-Stream Causality Reasoning Module, including Graph Convolutional Network, Graph Attention Network, HypergraphConv, and TransformerConv. While their overall performance is relatively comparable, the Graph Attention Network shows stronger capability in capturing event causal relations and achieves the highest F1 score.

4.6. Case study (RQ4)

In Fig. 6, we compare our baseline approach with the CogECI method based on instance results. The intra-sentence example illustrates that the baseline approach exhibits a bias toward common knowledge, leading to incorrect choices by the model. In contrast, CogECI correctly analyzes the grounded context "S7" and gets accurate conclusions. This demonstrates that even intra-sentence problems rely on contextual dependencies. When analyzing inter-sentence problems, supported by the correct contextual content, CogECI can derive accurate results. However, the baseline lacks adequate understanding of contextual information and fails to establish the correct association between "bombing" and "landed". Both examples demonstrate the effectiveness of our proposed method.

5. Conclusion

In this paper, we propose a novel **Co**ntext **G**rounded Document-level **E**vent **C**ausality **I**dentification (CogECI) Network to eliminate noisy context and the impact of bad language shortcuts. Specifically, we first utilize Large Language Models for their extensive commonsense knowledge and advanced semantic understanding capabilities to extract the grounded context pseudo-labels from the document that are semantically contributive to the causality reasoning of the event pairs. Then, we design a sentence graph module (SGM) to establish global sentence-level dependencies, addressing the limitations of traditional sliding window approaches that fail to capture global semantic relations across the document. In addition, the Dual-Stream Causality Reasoning Module performs event causality reasoning from inter-sentence and intra-sentence levels. Finally, the grounded context pseudo-labels are used for Context Grounding Learning to eliminate the bad influence of numerous redundant and noisy long contexts. Extensive experiments on two benchmark datasets show that CogECI not only outperforms previous state-of-the-art (SOTA) methods but also obtains a precise context grounding ability.

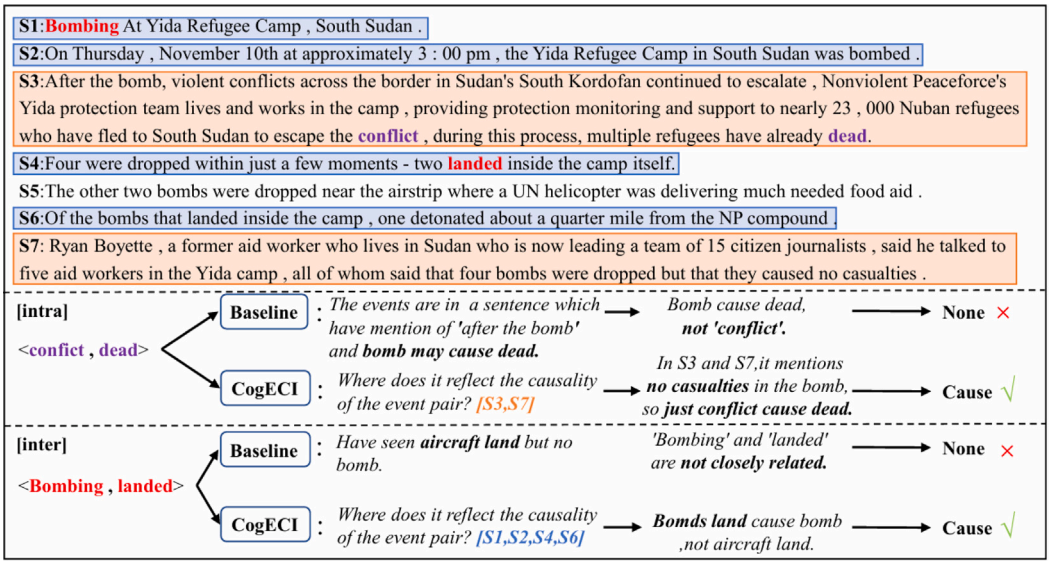


Fig. 6. The case study of CogECI. The baseline in this study is a relatively simple graph-based structure without Context Grounding loss and Sentence Graph Module. Additionally, the same graph reasoning module is used for intra-sentence and inter-sentence reasoning.

CRediT authorship contribution statement

Zefan Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology. **Xumeng Zhang:** Writing – review & editing, Visualization, Software, Conceptualization. **Shijie Jiang:** Writing – review & editing, Validation, Software. **Tian Bai:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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